

Program : Diploma in Communication and Computer Networking	
Course Code : 5323	Course Title: Fundamentals of Embedded System Design
Semester: 5	Credits: 4
Course Category: Programme core course	
Periods per week: 4	Periods per semester: 60

Course Objectives:

- Understand basics of Microcontrollers and functioning
- Understand Arduino and programming
- Understand interfacing of a microcontroller for communication.
- Understand the architecture of real time operating systems

Course Prerequisites:

Topic	Course code	Course name	Semester
Sound knowledge in Logic System Design and Computer organization & architecture.	3322	Logic Design	3
	3321	Computer Architecture and Organization	

Course Outcomes:

On completion of the course, the student will be able to:

CO	Description	Duration (Hours)	Cognitive Level
CO1	To Describe microprocessors functioning	15	Understanding
CO2	To Recognize the functioning of ARDUINO	16	Understanding
CO3	To familiarise the interfacing of ARDUINO	15	Understanding
CO4	To understand the concept of real time operating systems	12	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2		3	2			
CO2	2		3		2		
CO3	2		2	2			
CO4	2				3		2

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	To Describe microprocessors functioning		
M1.01	To understand Microprocessors	6	Understanding
M1.02	To understand interfacing of Microprocessors	9	Understanding
Contents: Introduction to Microprocessor: 8051 microprocessor-Overview of 8051-Principles of UART-Data Format-Syntax			
CO2	To Recognize the functioning of ARDUINO		
M2.01	To understand features of Arudino	8	Understanding
M2.02	To understand basic programming of 8051	8	Understanding
	Series Test –I	1	
Contents: Introduction to Arduino 8051 programming-Hardware overview-Configuration and installation-Arduino IDE- Sketch Overview Syntax-Sensor and actuator using Arduino			
CO3	To familiarise the interfacing of ARDUINO		
M3.01	To understand interfacing with IC	15	Understanding
Contents: Case study Arudino Case study-Interface of switches and LED-Interfacing with keyboard-Using buttons-Blink LED without using a delay function			

CO4	To understand the concept of real time operating systems		
M 4.01	To understand sensors and actuators in real time operating systems	12	Understanding
	Series Test – II	1	
Contents: Overview of RTOS Sensors-Different types of sensors-Introduction to RTOS-Different types of RTOS-Architecture-Case study-RTOS for control system			

Text / Reference

T/R	Book Title/Author
T1	Shibu.K.V, “Introduction to Embedded Systems”, TataMcgraw Hill,2009
T2	Peckol, “Embedded system Design”,JohnWiley&Sons,2010
T3	Lyla B.Das : Microprocessors and Microcontrollers, Pearson Education, India, 2011
R1	Rajkamal, ‘Embedded system-Architecture, Programming, Design’, TMH,2011
R2	Jorgen Staunstrup, Wayne Wolf ,Hardware / Software Co- Design Principles and Practice, Springer, 2009.
R3	Sam Siewert, V, Real-Time Embedded Components and Systems: With Linux and RTOS (Engineering), 2015
R4	Tanenbaum, Modern Operating Systems, 3/e, Pearson Edition, 2007.
R5	Wayne Wolf, Computers as Components: Principles of Embedded Computing System Design, 2/e, Kindle Publishers, 2005.

Online Resources

Sl.No	Website Link
1	https://onlinecourses.nptel.ac.in/noc21_ee18 https://www.udemy.com/course/mastering-microcontroller-with-peripheral-driver-development
2	